

L_1 pattern matching lower bound

Ohad Lipsky¹, Ely Porat^{*,2}

Department of Computer Science, Bar-Ilan University, Ramat-Gan 52900, Israel

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Abstract

Let a text string $T = t_0, \dots, t_{n-1}$ and a pattern string $P = p_0, \dots, p_{m-1}$, $t_i, p_j \in \mathbb{N}$ be given. In the *Approximate Pattern Matching in the L_1 metric* problem (L_1 -matching for short) the output is, for every text location i , the L_1 distance between the pattern and the length m substring of the text starting at i , i.e. $\sum_{j=0}^{m-1} |t_{i+j} - p_j|$. The *Less Than Matching* problem is that of finding all locations i of T where $t_{i+j} \geq p_j$, $j = 0, \dots, m-1$. The *String Matching with Mismatches* problem is that of finding the number of mismatches between the pattern and every length m substring of the text. For the three above problems, the fastest known deterministic solution is $O(n\sqrt{m \log m})$ time.

In this paper we show that the latter two problems can be linearly reduced to the problem of L_1 -matching.

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1. Introduction

Approximate matching is one of the fundamental problems in pattern matching. In approximate matching one defines a distance function between strings, and seeks for the distance between the pattern and every length m substring of the text. In the problem of *String Matching with Mismatches* [6] the hamming distance is used as distance function. Almost two decades ago Abrahamson [1] showed an $O(n\sqrt{m \log m})$ time algorithm for the problem, which is the fastest known so far.

Amir, Lewenstein and Porat [3] gave an $O(n\sqrt{k \log k})$ time algorithm for the case that we can discard locations with distance greater than a given bound k . However, in the general case $k = m$ and it is not better than Abrahamson's result. More than that, Indyk [9] showed that improvement over the Abrahamson algorithm to $O(nm^c)$ time will yield an $O(n^{2+2c})$ time for boolean matrix multiplication.

The number of mismatches is an important measure as part of an *edit distance* between typed strings. Advances in Multimedia, Digital Libraries and Computational Biology have shown that a much more generalized theoretical basis of string matching could be of tremendous benefit [16,15]. In computer vision, for example, it does not make sense to say that a pattern pixel with a close grey level to the text pixel should generate the same error penalty as, say, a white pattern pixel being matched to a black text pixel. Similarly in biology,

* Corresponding author. Tel.: +972 3 531 7620.

E-mail addresses: ohadlipsky@yahoo.com (O. Lipsky), porately@cs.biu.ac.il (E. Porat).

¹ Tel.: +972 3 531 8408.

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the energy level for bonding different proteins is different, suggesting that not all “mismatches” should be counted equally. In various other applications, such as earthquake prediction [14], stock market analysis [13], and music retrieval [17], the distance measure used is the Minkowsky L_1 norm. The L_1 norm is of particular importance also due to the fact that strings can be embedded in the L_1 space and the distance between their associated vectors approximates the *edit distance with moves* of the strings [8]. The problem of L_1 -matching was solved in $O(n\sqrt{m\log m})$ time in [12,4,7]. The solution of [12] was further extended to solve closest pair problems in [10]; it was also the base for other pattern matching algorithms in [11]. We believe it is hard to show a better time for the L_1 -matching.

Our contribution. We show here that *String Matching with Mismatches* can be linearly reduced to L_1 -matching. We further show that the problem of *Less Than Matching* can also be linearly reduced to L_1 -matching.

The problem of *Less Than Matching* was presented in [2] and solved in $O(n\sqrt{m\log m})$. It is still open problem whether that time is optimal.

2. Preliminaries

Let $A = a_0, \dots, a_{m-1}$ and $B = b_0, \dots, b_{m-1}$ be two string of numbers. The L_1 distance between A and B , denoted by $L1(A, B)$ is defined by $L1(A, B) = \sum_{j=0}^{m-1} |a_j - b_j|$.

Formally define the **Approximate Pattern Matching in the L_1 metric** problem:

Input: Text $T = t_0, \dots, t_{n-1}$, pattern $P = p_0, \dots, p_{m-1}$, where $t_i, p_j \in \mathbb{N} \forall i, j$.

Output: $l_0, l_1, \dots, l_{n-m}$ where $l_i = L1(t_i \dots t_{i+m-1}, P)$ for every $i \in [0, n - m]$.

The problem of **String Matching with Mismatches** is the problem of counting the mismatches between the pattern and the text, for every alignment possible. Formally:

Input: Text $T = t_0, t_1, \dots, t_{n-1}$ and pattern $P = p_0, p_1, \dots, p_{m-1}$, where $t_i, p_j \in \Sigma$.

Output: $h_0, \dots, h_{n-m}$ where $h_i = \sum_{j=0}^{m-1} neq(t_{i+j}, p_j)$ for every $i \in [0, n - m]$ and where

$$neq(x, y) = \begin{cases} 1, & x \neq y, \\ 0, & x = y. \end{cases}$$

The problem of **Less than Matching** is defined as:

Input: text $T = t_0, \dots, t_{n-1}$, pattern $P = p_0, \dots, p_{m-1}$ where $t_i, p_j \in \mathbb{N}$.

Output: All locations i in T $i \in [0, n - m]$ s.t. $t_{i+j} \geq p_j \forall j = 0, \dots, m - 1$.

3. Reduction from string matching with mismatches

We use the following key observation:

Lemma 1. For any pair of integers x, y

$$\begin{aligned} & |x - y + 1| + |x - y - 1| - 2|x - y| \\ &= \begin{cases} 0, & x \neq y, \\ 2, & x = y. \end{cases} \end{aligned}$$

The reduction. The input is a text $T = t_0 t_1 \dots t_{n-1}$ and a pattern $P = p_0 p_1 \dots p_{m-1}$ both string of symbols from some alphabet Σ . We can assume that $\Sigma = \{1, 2, \dots, |\Sigma|\}$. If not, just change T and P using any bijective function from Σ to $\{1, 2, \dots, |\Sigma|\}$. This can be done in linear time.

1. Construct P^- to be $p_0 - 1, p_1 - 1, \dots, p_{m-1} - 1$ and P^+ to be $p_0 + 1, p_1 + 1, \dots, p_{m-1} + 1$.
2. Let $l_0, l_1, \dots, l_{n-m}$ be the output of L_1 -matching of T and P ,
 $l_0^+, l_1^+, \dots, l_{n-m}^+$ be the output of L_1 -matching of T and P^+ and
 $l_0^-, l_1^-, \dots, l_{n-m}^-$ be the output of L_1 -matching of T and P^- .
3. For every $i \in [0, n - m]$:

$$m_i \leftarrow \frac{1}{2}(l_i^+ + l_i^- - 2l_i).$$

$$h_i \leftarrow m - m_i.$$

It follows from the observation that m_i equals to the number of matches at location i and then clearly h_i is the number of mismatches at location i . The reduction is linear.

4. Reduction from less than matching

Lemma 2. For any pair of integers x, y

$$|x - y + 1| - |x - y| = \begin{cases} +1, & x \geq y, \\ -1, & x < y. \end{cases}$$

The reduction. The input is text $T = t_0 t_1 \dots t_{n-1}$ and pattern $P = p_0 p_1 \dots p_{m-1}$ both strings of natural numbers. Take the following steps:

1. Run L_1 -matching with T and P , output will be $l_0, \dots, l_{n-m}$.
2. Run L_1 -matching with T^+ and P , where $T^+ = t_0 + 1, t_1 + 1, \dots, t_{n-1} + 1$. The output will be $l_0^+, l_1^+, \dots, l_{n-m}^+$.
3. For every $i \in [0, n - m]$:
If $l_i^+ - l_i = m$ then output location i as a *match*.

Clearly, the reduction is linear. At location i , every pair of numbers t_{i+j} and p_j can either add 1 to $l_i^+ - l_i$ or subtract 1 from $l_i^+ - l_i$, it follows that if $l_i^+ - l_i = m$ all pairs added 1, which mean that for all $j = 0, \dots, m - 1$ it holds that $t_{i+j} \geq p_j$, which mean we have a match. It is easily seen that if there is a match at location i it will hold that $l_i^+ - l_i = m$.

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